

Customer Churn Analytics

Executive Analytics Capstone

From customer behavior to explainable retention prioritization

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Business Problem

- Customer churn can reduce recurring revenue.
- Different customers may require different retention actions.
- The business needs a simple way to identify customers requiring attention.
- Customer behavior can be converted into actionable segments and risk signals.

Decision Question

Which customer groups should receive retention attention, and why?

Executive Snapshot

TOTAL CUSTOMERS
15

OBSERVED CHURN
Use your Colab value

OBSERVED CHURN RATE
Use your Colab value

HIGH-RISK CUSTOMERS
Use your Colab value

HIGH-RISK MONTHLY CHARGES
Use your Colab value

Insight 1 — Contract Type and Churn

- Compare observed churn rates across Month-to-Month, One Year, and Two Year contracts.
- Contract type is used as a prioritization signal in the model.

Important: observed differences do not establish that contract type causes churn.

Insight 2 — Tenure and Support Activity

- Compare average tenure between churned and non-churned customers.
- Compare average support tickets between churned and non-churned customers.

Shorter tenure and higher support activity are treated as prioritization signals, not proven causes of churn.

Insight 3 — Customer Segmentation

Loyal & Stable

Maintain relationship and routine engagement.

New & High Support

Proactive onboarding and support review.

Newer Customers

Focus on early engagement.

Established Customers

Monitor and maintain relationship.

Insight 4 — Explainable Churn-Risk Score

SHORT TENURE +2

MONTH-TO-MONTH CONTRACT +2

HIGHER SUPPORT ACTIVITY +1

Maximum Score = 5

0–1 Low Risk

2–3 Medium Risk

4–5 High Risk

The score is a prioritization tool, not a prediction that an individual customer will churn.

Insight 5 — Recommended Actions & Impact

1. Prioritize higher-risk customers for retention review.
2. Strengthen onboarding for newer customers with higher support activity.
3. Monitor month-to-month customers through appropriate engagement initiatives.
4. Track repeated support activity as an early review signal.

Illustrative impact: apply a 10–20% retention scenario to monthly charges associated with the high-risk group.

This is a scenario estimate, not a forecast. Replace the impact range with the values calculated in Colab.

Limitations & Next Steps

LIMITATIONS

- Dataset contains only 15 customers.
- Results should not be generalized to a larger population.
- Risk score uses simple business rules.
- Observed churn was used for validation, not as a scoring input.
- Impact calculation is scenario-based, not a forecast.

NEXT STEPS

- Test the model on a larger historical dataset.
- Monitor retention outcomes.
- Recalibrate thresholds using additional evidence.
- Compare risk scores with future churn outcomes.